

TMUA PRACTICE TEST SERIES

TMUA Topic Test 3

Calculus

- 20 multiple-choice questions
- Recommended time: 75 minutes
- Non-calculator
- Paper 1 styled topic revision set

SOLUTION BOOK

ThrivingScholars

TMUA Topic Test 3 — Calculus

Complete, independently checked worked solutions for the 20-question TMUA Paper 1 styled calculus topic test.

TMUA Topic Test 3 answer key

Q1	A	Q2	C	Q3	C	Q4	D	Q5	A	Q6	B	Q7	F	Q8	C	Q9	A	Q10	B
Q11	D	Q12	A	Q13	D	Q14	B	Q15	D	Q16	F	Q17	A	Q18	D	Q19	D	Q20	E

WORKED SOLUTIONS

Integration, differentiation and optimisation

Questions 1–6 · substitution, structural differentiation, areas, increasing functions, optimisation and basic differential equations.

Q1

Integration & Substitution

Answer A

Question 1

Compute the integral

$$\int_0^1 \frac{x-4}{\sqrt{x}+2} dx.$$

A $-\frac{4}{3}$ **B** -3 **C** -1 **D** $\frac{2}{3}$ **E** 1

WORKED SOLUTION

Let $t = \sqrt{x}$. Then $x = t^2$, $dx = 2t dt$, and the limits remain 0 to 1.

$$\begin{aligned}
 I &= \int_0^1 \frac{t^2 - 4}{t + 2} 2t \, dt \\
 &= \int_0^1 2t(t - 2) \, dt \\
 &= \int_0^1 (2t^2 - 4t) \, dt \\
 &= \left[\frac{2}{3}t^3 - 2t^2 \right]_0^1 = -\frac{4}{3}.
 \end{aligned}$$

Therefore the answer is **A**.

Q2

Differentiation & Structure

Answer C

Question 2

Given

$$f(x) = \left(9x^2 + 12 + \frac{4}{x^2} \right)^{1/2}$$

and

$$\frac{d^n f}{dx^n}(2) = -\frac{3}{4},$$

find n .

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

The expression inside the square root is a perfect square:

$$9x^2 + 12 + \frac{4}{x^2} = \left(3x + \frac{2}{x} \right)^2.$$

Near $x = 2$, the quantity $3x + 2/x$ is positive, so

$$f(x) = 3x + \frac{2}{x}.$$

Differentiate:

$$f'(x) = 3 - \frac{2}{x^2}, \quad f''(x) = \frac{4}{x^3}, \quad f^{(3)}(x) = -\frac{12}{x^4}.$$

Thus

$$f^{(3)}(2) = -\frac{12}{16} = -\frac{3}{4}.$$

Hence $n = 3$, so the answer is **C**.

Q3

Area Between Curves

Answer C

Question 3

For $p > 0$, find the area enclosed by the curves

$$y = px^2 \quad \text{and} \quad x = py^2.$$

A $\frac{1}{3p}$

B $3p$

C $\frac{1}{3p^2}$

D $\frac{1}{2}p^2$

E $\frac{1}{6p^2}$

WORKED SOLUTION

From $x = py^2$, the upper branch in the first quadrant is

$$y = \sqrt{\frac{x}{p}}.$$

At an intersection with $y = px^2$,

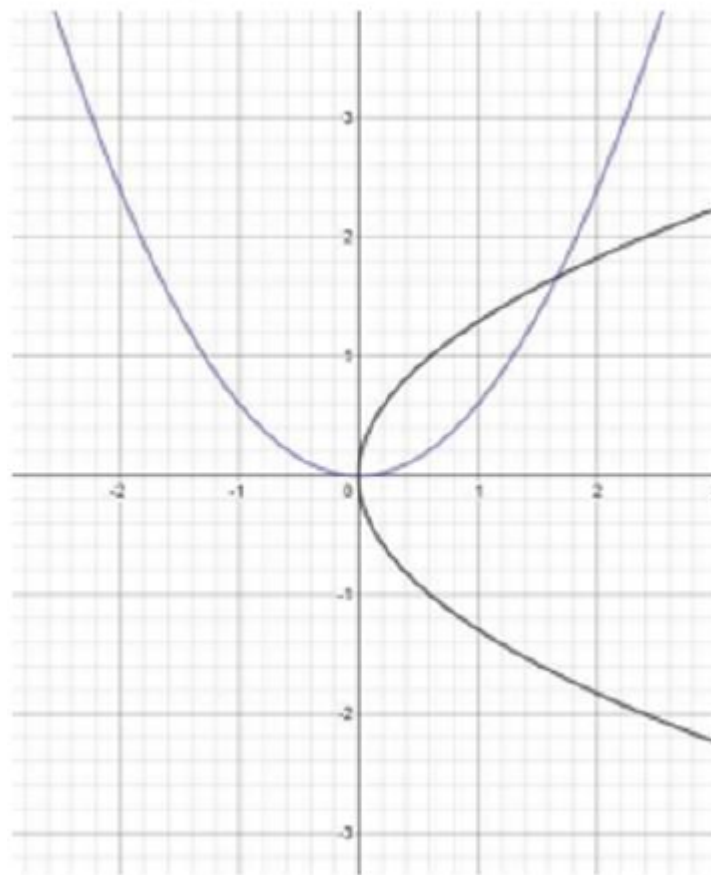
$$x = p(px^2)^2 = p^3x^4,$$

so $x = 0$ or $x = 1/p$. The corresponding non-zero point is $(1/p, 1/p)$.

Therefore the enclosed area is

$$\begin{aligned} A &= \int_0^{1/p} \left(\sqrt{\frac{x}{p}} - px^2 \right) dx \\ &= \left[\frac{2x^{3/2}}{3\sqrt{p}} - \frac{px^3}{3} \right]_0^{1/p} \\ &= \frac{2}{3p^2} - \frac{1}{3p^2} = \frac{1}{3p^2}. \end{aligned}$$

The answer is **C**.



Original supporting graph URL retained from the supplied solution PDF, with an embedded fallback for offline use.

Q4

Increasing Functions

Answer D

Question 4

What is the complete set of values for which

$$\frac{x^2 + 2x}{\sqrt{x^3}}$$

is increasing?

A $x > 3$

B $x < 2$

C $2 < x < 4$

D $x > 2$

E $0 < x < 2$

WORKED SOLUTION

The denominator requires $x > 0$. Rewrite the function as

$$x^{1/2} + 2x^{-1/2}.$$

Its derivative is

$$\frac{1}{2\sqrt{x}} - \frac{1}{x^{3/2}} = \frac{x-2}{2x^{3/2}}.$$

For $x > 0$, the denominator is positive, so the derivative is positive precisely when $x > 2$.

The answer is **D**.

Q5

Optimisation & Distance

Answer A

Question 5

A point P lies on the part of the parabola $y = x^2$ in the first quadrant. What is the shortest possible distance from P to the point $(0, 3)$?

A $\frac{\sqrt{11}}{2}$

B $\frac{3}{2}$

C $\frac{\sqrt{5}}{2}$

D $\frac{11}{4}$

E $\sqrt{3}$

WORKED SOLUTION

Write $P = (x, x^2)$, where $x \geq 0$. The square of the distance to $(0, 3)$ is

$$D^2 = x^2 + (x^2 - 3)^2 = x^4 - 5x^2 + 9.$$

Minimising D is equivalent to minimising D^2 . Differentiate:

$$\frac{d}{dx}(D^2) = 4x^3 - 10x = 2x(2x^2 - 5).$$

The non-zero stationary point in the first quadrant is $x^2 = 5/2$. There,

$$D^2 = \frac{5}{2} + \left(\frac{5}{2} - 3\right)^2 = \frac{5}{2} + \frac{1}{4} = \frac{11}{4}.$$

Hence

$$D_{\min} = \frac{\sqrt{11}}{2}.$$

The answer is **A**.

Question 6

Given that

$$\frac{dV}{dt} = (1 + t)^4$$

and $V(1) = 5$, what is $V(2)$?

A $\frac{174}{5}$

B $\frac{236}{5}$

C $\frac{112}{3}$

D $\frac{89}{4}$

E $\frac{17}{32}$

WORKED SOLUTION

Use the definite change in V :

$$V(2) - V(1) = \int_1^2 (1 + t)^4 dt = \left[\frac{(1 + t)^5}{5} \right]_1^2 = \frac{3^5 - 2^5}{5} = \frac{211}{5}.$$

Therefore

$$V(2) = 5 + \frac{211}{5} = \frac{236}{5}.$$

The answer is **B**.

Stationary points, symmetry and numerical methods

Questions 7–13 · parameterised stationary points, definite-integral comparisons, symmetry, turning points, the trapezium rule and cubic root conditions.

Q7

Stationary Points & Parameters

Answer F

Question 7

Let

$$y = \left(\frac{1}{3}k^3 + 1\right)x^3 - (2k^2 + 3)x^2 + (31 - 25k)x + 43.$$

The curve has a turning point that is a maximum at $x = 1$, precisely for:

A $-4 < k < 7$

B $k = 1$ or 7

C $k > 7$

D $k = -4$ or 7

E $k < 1$

F $k = -4$ or 1

G $k < -4$

H $k = -4, 1$ or 7

WORKED SOLUTION

For a turning point at $x = 1$, require $y'(1) = 0$.

$$y'(1) = k^3 - 4k^2 - 25k + 28 = (k - 1)(k - 7)(k + 4).$$

Thus the possible values are

$$k = -4, 1, 7.$$

Now

$$y''(1) = 2k^3 - 4k^2 = 2k^2(k - 2).$$

A maximum requires $y''(1) < 0$, so among the three candidates only $k = -4$ and $k = 1$ work.

The answer is **F**.

Question 8

Which of the following integrals has the largest value?

A $\int_0^{\pi/3} \cos\left(\frac{\pi}{2} - x\right) dx$

B $\int_{\pi/3}^{2\pi/3} \cos^2 x dx$

C $\int_{2\pi/3}^{\pi} -\sin\left(\frac{\pi}{2} - x\right) dx$

D $\int_0^{\pi} \sin\left(\frac{\pi}{2} - x\right) dx$

WORKED SOLUTION

Simplify each integrand:

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x, \quad -\sin\left(\frac{\pi}{2} - x\right) = -\cos x.$$

Therefore

$$A = \int_0^{\pi/3} \sin x dx = \frac{1}{2},$$

$$B = \int_{\pi/3}^{2\pi/3} \cos^2 x dx = \frac{\pi}{6} - \frac{\sqrt{3}}{4},$$

$$C = \int_{2\pi/3}^{\pi} -\cos x dx = \frac{\sqrt{3}}{2}, \quad D = \int_0^{\pi} \cos x dx = 0.$$

Since $\sqrt{3}/2 > 1/2$, and B is much smaller than $1/2$, the largest value is C .

The answer is **C**.

Question 9

Find the value of

$$\int_{-1}^1 \frac{x(x^2 - 2)^2}{x^2 + 1} dx.$$

A 0

B 1

C 2

D 3

WORKED SOLUTION

Let

$$f(x) = \frac{x(x^2 - 2)^2}{x^2 + 1}.$$

The numerator contains one factor of x multiplied by a function of x^2 , while the denominator is also a function of x^2 . Hence

$$f(-x) = -f(x).$$

So f is odd and continuous on $[-1, 1]$. Therefore

$$\int_{-1}^1 f(x) dx = 0.$$

The answer is **A**.

Q10

Turning Points & Distance

Answer B

Question 10

A curve has equation

$$y = (2a - x)(2a + x),$$

and a second curve has equation

$$y = (2a - x)(3a + x),$$

where $a \neq 0$. The distance between the maximum points of the two curves is $\sqrt{10}/3$. Find the possible values of a .

A 0

B $\pm \frac{2}{3}$ C $\pm \frac{1}{2}$ D $\pm \frac{\sqrt{10}}{3}$ E ± 1 F $\pm \frac{2\sqrt{5}}{5}$

WORKED SOLUTION

The first curve is

$$y = 4a^2 - x^2,$$

so its maximum point is

$$P = (0, 4a^2).$$

The second curve is

$$y = -x^2 - ax + 6a^2,$$

whose maximum occurs at $x = -a/2$. Substitution gives

$$Q = \left(-\frac{a}{2}, \frac{25a^2}{4}\right).$$

Hence

$$PQ^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{9a^2}{4}\right)^2 = \frac{a^2}{4} + \frac{81a^4}{16}.$$

Since $PQ = \sqrt{10}/3$,

$$\frac{a^2}{4} + \frac{81a^4}{16} = \frac{10}{9}.$$

Let $u = a^2$. Then

$$729u^2 + 36u - 160 = 0 = (9u - 4)(81u + 40).$$

As $u \geq 0$, $u = 4/9$. Therefore

$$a = \pm \frac{2}{3}.$$

The answer is **B**.

Q11

Properties of Definite Integrals

Answer D

Question 11

A function f satisfies

$$\int_0^4 f(x) dx = 10, \quad \int_{-1}^4 f(x) dx = 13, \quad \int_{-4}^3 f(-x) dx = 2.$$

Find

$$\int_{-3}^0 f(x) dx.$$

A 2

B 3

C 5

D -8

E 11

F 12

WORKED SOLUTION

First,

$$\int_{-1}^0 f(x) dx = \int_{-1}^4 f(x) dx - \int_0^4 f(x) dx = 13 - 10 = 3.$$

For the third integral, use $u = -x$. As x runs from -4 to 3 , u runs from 4 to -3 , so

$$\int_{-4}^3 f(-x) dx = \int_{-3}^4 f(u) du = 2.$$

Therefore

$$\int_{-3}^0 f(x) dx = \int_{-3}^4 f(x) dx - \int_0^4 f(x) dx = 2 - 10 = -8.$$

The answer is **D**.

Q12

Trapezium Rule

Answer A

Question 12

The trapezium-rule approximation to

$$\int_0^6 |x(x-3)(x-6)| dx$$

using four trapezia of equal width is:

A $\frac{3^5}{2^3}$ B $\left(\frac{3}{2}\right)^4$ C $\frac{3^5}{4}$ D $\frac{3^3}{2^5}$ E $\left(\frac{3}{2}\right)^5$

F 30

WORKED SOLUTION

Four trapezia on $[0, 6]$ have width

$$h = \frac{6}{4} = \frac{3}{2}.$$

For $F(x) = |x(x-3)(x-6)|$, the five ordinates are

$$F(0) = 0, \quad F\left(\frac{3}{2}\right) = \frac{81}{8}, \quad F(3) = 0, \quad F\left(\frac{9}{2}\right) = \frac{81}{8}, \quad F(6) = 0.$$

The trapezium rule gives

$$\frac{h}{2}[y_0 + 2y_1 + 2y_2 + 2y_3 + y_4] = \frac{3}{4}\left(4 \cdot \frac{81}{8}\right) = \frac{243}{8} = \frac{3^5}{2^3}.$$

The answer is **A**.

Q13

Cubic Graphs & Parameters

Answer D

Question 13

Find the complete set of values of k for which

$$2x^3 - 3x^2 - 36x + k = 0$$

has three distinct real solutions.

A $k < 44$

B $k < 81$

C $-81 < k < 44$

D $-44 < k < 81$

E $k > 44$

F $k > 81$

WORKED SOLUTION

Let

$$h(x) = 2x^3 - 3x^2 - 36x.$$

Then

$$h'(x) = 6x^2 - 6x - 36 = 6(x+2)(x-3),$$

so the turning points occur at $x = -2$ and $x = 3$. Their values are

$$h(-2) = 44, \quad h(3) = -81.$$

The original equation is $h(x) = -k$. It has three distinct real roots exactly when the horizontal line $y = -k$ lies strictly between the turning-point values:

$$-81 < -k < 44.$$

Multiplying by -1 reverses the inequalities:

$$-44 < k < 81.$$

The answer is **D**.

Advanced integration and calculus reasoning

Questions 14–20 · initial conditions, modulus integrals, symmetry, differential equations, constrained optimisation, rational indices and the Fundamental Theorem of Calculus.

Q14

Differential Equations

Answer B**Question 14**

A function y satisfies

$$\frac{dy}{dx} = 7 + \frac{3}{x}, \quad x > 0.$$

Its general solution is written as

$$y = 7x + 3 \ln x + C.$$

Given that $y = 5$ when $x = 3$, find C .

A $16 + 3 \ln 3$ **B** $-16 - 3 \ln 3$ **C** $-2 - 3 \ln 3$ **D** $-16 + 3 \ln 3$ **E** $21 + 3 \ln 3$ **F** 5 **WORKED SOLUTION**

Substitute the initial condition $(x, y) = (3, 5)$ into

$$y = 7x + 3 \ln x + C.$$

This gives

$$5 = 21 + 3 \ln 3 + C.$$

Hence

$$C = 5 - 21 - 3 \ln 3 = -16 - 3 \ln 3.$$

The answer is **B**.

Question 15

Find the value of

$$\int_0^2 |x-1| (3\sqrt{x} - x\sqrt{x}) dx.$$

A $\frac{48-4\sqrt{2}}{35}$

B $\frac{4\sqrt{2}}{35}$

C $\frac{24}{35}$

D $\frac{48+4\sqrt{2}}{35}$

E $\frac{24+4\sqrt{2}}{35}$

F $\frac{24-4\sqrt{2}}{35}$

WORKED SOLUTION

Split the integral at $x = 1$:

$$I = \int_0^1 (1-x)(3x^{1/2} - x^{3/2}) dx + \int_1^2 (x-1)(3x^{1/2} - x^{3/2}) dx.$$

Let

$$G(x) = (x-1)(3x^{1/2} - x^{3/2}).$$

An antiderivative of G is

$$F(x) = -\frac{2}{7}x^{7/2} + \frac{8}{5}x^{5/2} - 2x^{3/2}.$$

The first integrand is $-G(x)$, so

$$I = -[F(x)]_0^1 + [F(x)]_1^2 = F(2) - 2F(1).$$

Now

$$F(1) = -\frac{24}{35}, \quad F(2) = \frac{4\sqrt{2}}{35}.$$

Therefore

$$I = \frac{4\sqrt{2}}{35} + \frac{48}{35} = \frac{48+4\sqrt{2}}{35}.$$

The answer is **D**.

Question 16

For a function f ,

$$\left(\int_0^2 f(x) dx\right) \left(\int_0^1 f(x) dx + 4\right) = 24.$$

Also, $f(x+1) = f(1-x)$. Which of the following could be the value of

$$\int_1^2 f(x) dx?$$

I. -6

II. 8

III. 2

A None

B I only

C II only

D III only

E I & II only

F I & III only

G II & III only

H All three

WORKED SOLUTION

Let

$$A = \int_0^1 f(x) dx, \quad B = \int_1^2 f(x) dx.$$

Using $f(x+1) = f(1-x)$ and integrating for $0 \leq x \leq 1$,

$$B = \int_0^1 f(x+1) dx = \int_0^1 f(1-x) dx = A.$$

Thus $\int_0^2 f = 2A$, and the given condition becomes

$$2A(A+4) = 24.$$

So

$$A^2 + 4A - 12 = 0 \implies (A+6)(A-2) = 0.$$

Hence $A = B = -6$ or 2 . Statements I and III are possible.

Q17**Differential Equations & Trigonometry****Answer A****Question 17**

Given that

$$\frac{dy}{dx} = \frac{3}{2} \left(\frac{x^{7/3} - x^{5/3}}{x^{1/3} - 1} + \frac{x^{7/3} - x^{5/3}}{x^{1/3} + 1} \right), \quad x > 1,$$

and $y = 4$ when $x = 2$, find y when $x = 5$.

A 121**B** 117**C** 125**D** 113**E** 129**F** 81**G** $\frac{243}{2}$ **WORKED SOLUTION**

Let $u = x^{1/3}$. Then

$$x^{7/3} - x^{5/3} = x^{5/3}(x^{2/3} - 1) = u^5(u - 1)(u + 1).$$

Therefore

$$\frac{x^{7/3} - x^{5/3}}{x^{1/3} - 1} = u^5(u + 1), \quad \frac{x^{7/3} - x^{5/3}}{x^{1/3} + 1} = u^5(u - 1).$$

Adding the two fractions gives

$$u^5(u + 1) + u^5(u - 1) = 2u^6 = 2x^2.$$

Hence

$$\frac{dy}{dx} = \frac{3}{2}(2x^2) = 3x^2,$$

so

$$y = x^3 + C.$$

Using $y = 4$ when $x = 2$,

$$4 = 8 + C \implies C = -4.$$

Thus, when $x = 5$,

$$y = 5^3 - 4 = 125 - 4 = \boxed{121}.$$

The correct answer is A.

Q18

Constrained Integration

Answer D

Question 18

Given that

$$\int_0^4 (x^3 + ax^2 + bx + 1)dx = 0,$$

where $a \leq 0$ and $b \leq 0$, find the maximum value of

$$\int_0^3 (x^3 + ax^2 + bx + 1)dx.$$

A -15

B 15

C 100

D $-\frac{87}{16}$

E $\frac{123}{2}$

F $-\frac{123}{2}$

WORKED SOLUTION

The first condition gives

$$68 + \frac{64a}{3} + 8b = 0,$$

so

$$b = -\frac{17}{2} - \frac{8a}{3}.$$

Because $b \leq 0$,

$$-\frac{17}{2} - \frac{8a}{3} \leq 0 \implies a \geq -\frac{51}{16}.$$

Together with $a \leq 0$, this gives

$$-\frac{51}{16} \leq a \leq 0.$$

Now

$$\begin{aligned}\int_0^3 (x^3 + ax^2 + bx + 1)dx &= \frac{93}{4} + 9a + \frac{9b}{2} \\ &= -15 - 3a.\end{aligned}$$

This is largest when a is smallest, namely $a = -51/16$. Therefore the maximum is

$$-15 + \frac{153}{16} = -\frac{87}{16}.$$

The answer is **D**.

Q19

Rational Indices & Differentiation

Answer D

Question 19

Let

$$f(x) = \frac{(x^2 - \frac{1}{x})^2}{x^{1/3}}.$$

It is also given that

$$f'(x) = \frac{x^{2/3}}{3}(ax^p + bx^q + cx^r).$$

Find $abc + pqr$.

A 388

B $\frac{520}{3}$

C 288

D 316

E 300

F -80

WORKED SOLUTION

First expand and simplify:

$$\begin{aligned}f(x) &= \frac{x^4 - 2x + x^{-2}}{x^{1/3}} \\ &= x^{11/3} - 2x^{2/3} + x^{-7/3}.\end{aligned}$$

Differentiate:

$$f'(x) = \frac{11}{3}x^{8/3} - \frac{4}{3}x^{-1/3} - \frac{7}{3}x^{-10/3}.$$

Factor out $x^{2/3}/3$:

$$f'(x) = \frac{x^{2/3}}{3}(11x^2 - 4x^{-1} - 7x^{-4}).$$

Thus

$$a = 11, b = -4, c = -7, \quad p = 2, q = -1, r = -4.$$

Therefore

$$abc + pqr = 11(-4)(-7) + 2(-1)(-4) = 308 + 8 = 316.$$

The answer is **D**.

Q20

Fundamental Theorem & Modulus

Answer E

Question 20

Consider

$$g(x) = \int_0^x (t - p)f(t) dt, \quad 0 \leq p \leq 7.$$

Find

$$\int_0^7 |x - p|f(x) dx.$$

A $g(7) - g(0)$

B $g(7) - g(p)$

C $g(0) - pg(7)$

D $2g(0) - g(7) + g(p)$

E $g(7) + g(0) - 2g(p)$

F $g(p) + g(7) - 2g(0)$

WORKED SOLUTION

By the Fundamental Theorem of Calculus,

$$g'(x) = (x - p)f(x).$$

Split the modulus integral at $x = p$:

$$\begin{aligned} \int_0^7 |x - p|f(x)dx &= \int_0^p (p - x)f(x)dx + \int_p^7 (x - p)f(x)dx \\ &= -\int_0^p (x - p)f(x)dx + \int_p^7 (x - p)f(x)dx \\ &= -[g(p) - g(0)] + [g(7) - g(p)] \\ &= g(7) + g(0) - 2g(p). \end{aligned}$$

The answer is **E**.